

Exam Date & Time: 02-Jun-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FOURTH SEMESTER B.TECH. DEGREE EXAMINATIONS - MAY/JUNE 2023

Principles of soft computing [CSE-4305]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Distinguish between Supervised and Unsupervised learning algorithm. (2)
 - A)
 - B) A single input, single output neuron has a weight of 5 and a bias of -3. What is the net input to the transfer function if it is given an input of 1? What will be the output if the transfer function is sigmoid activation function? (3)
 - C) Explain Error correction learning method in detail with suitable example. (5)
- 2) Explain and draw the architecture of a Convolutional Neural networks and explain its operation in terms of each layer. (5)
 - A)
 - B) Explain Gradient descent algorithm in detailed steps with suitable example. (5)
- 3) Draw the architecture of Associative memory network architecture and explain its operation. Mention its advantages and applications. (5)
 - A)
 - B) Explain operation involved in Kohonen network. Suppose, for instance that the 2-dimensional input vector X is presented to the three-neuron Kohonen network and W_1 , W_2 and W_3 are initial weight vectors. Find the winning neuron and updated weights: $X = [0.52, 0.12]$, $W_1 = [0.27, 0.81]$, $W_2 = [0.42, 0.70]$, $W_3 = [0.43, 0.21]$. (5)
- 4) Distinguish between Crisp set and Fuzzy set. (2)
 - A)
 - B) Illustrate different fuzzy membership functions used in fuzzy logic. (3)
 - C) Design a fuzzy based model for automation of Anti-lock braking system. Mention input profiles, output profiles and rules. (5)

- 5) Illustrate advantages of Genetic algorithms. (2)
- A)
- B) Illustrate different mutation operations in Genetic algorithms. Consider a subset of genes (01**234567**89), apply inverse mutation operator for the bolded bits and represent a new representation of genes. (3)
- C) Explain steps involved in the working of Genetic algorithms using flowchart with help of suitable example. (5)

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